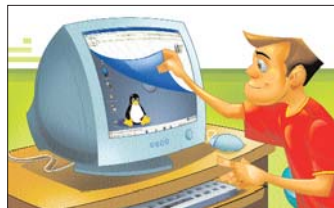




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Digital Business

How Technology Can Help Your Organisation

Tech Careers

Intelligence Inside

Embedded systems are everywhere. Think you can make everyday devices more useable—and perhaps come up with new ones?

Samir Makwana

India is fast emerging as a global technology hub: various multinational design labs, R&D houses, and product-centric start-ups are cropping up. In recent years, India's core competencies in software services, design, and R&D have attracted a lot of outsourced jobs. The innovations happening in the course of product design and developing software have put us on the global tech map.

Now, when you think of product design, embedded systems immediately come to mind. They're everywhere—in products in all categories.

We use smart devices and electronic equipment for various tasks, from enabling a smart door lock to watching a DVD on our home theatre. Smartphones, laptops, consumer electronics, automotive electronics, medical equipment, industrial machines, and, in fact, anything with a digital inter-

face, have their computing power in the form of microcontrollers or microprocessors on the integrated circuits "embedded" in them.

Certain embedded systems in products such as cell phones and PDAs have operating systems (such as Windows Mobile, Palm OS, and Symbian) specifically developed to perform various application functions besides the basic task of communication. Embedded systems in products such as industrial automated machines, automotive electronics, consumer electronics, etc. that are developed to perform a pre-defined task have a logical program written to the hardware. But what exactly are embedded systems?

In brief, embedded systems are a combination of hardware and software with some mechanical or other parts added, which perform discrete, limited functions. A program for performing a pre-defined function is written permanently onto the hardware such as a ROM chip.

For the hardware of the embedded system, microprocessors and microcontrollers of different architectures are used. Also, Real-Time Operating Systems (RTOSes) are used by software programmers to write programs for the hardware interfaces.

Illustration Pravin Warthekar

